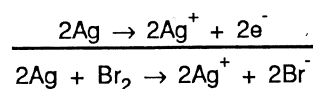
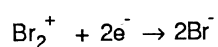
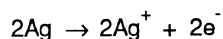
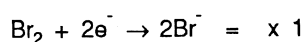
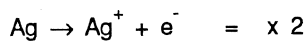
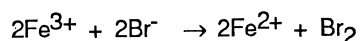


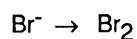
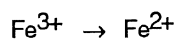
- (b) To determine the overall cell reaction, multiply the half-cell equations by factors which balance the electrons in the two equations and add the resulting equations.



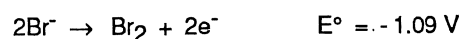
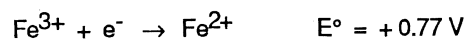
2. Predict whether the reaction below would occur in aqueous solution in which all concentrations are 1 mol L^{-1} .



- (a) Identify half-reactions involved



- (b) Write half-cell equations and reduction potentials



- (c) Calculate cell voltage

$$\begin{aligned} E^\circ &= +0.77 - 1.09 \\ &= -0.32 \text{ V} \end{aligned}$$

- (d) Since E° is negative the reaction will not occur. If a reaction has a positive E° it could occur as written.

SET 19

1. Calculate the standard cell voltages and write the overall chemical reactions for cells which consist of the following half-cells
 - (a) Zn^{2+}/Zn and Sn^{2+}/Sn
 - (b) Cr^{3+}/Cr and Ag^+/Ag
 - (c) Hg^{2+}/Hg and Cu^{2+}/Cu
 - (d) Mg^{2+}/Mg and Cu^{2+}/Cu
 - (e) Mg^{2+}/Mg and Ag^+/Ag
 - (f) $\text{Fe}^{3+}/\text{Fe}^{2+}$ and $\text{Cr}_2\text{O}_7^{2-}/\text{Cr}^{3+}$ ($\text{Cr}_2\text{O}_7^{2-}$ is acidified)
 - (g) Cl_2/Cl^- and I_2/I^-

Chemistry Problems

2. Predict whether the following reactions could occur under standard conditions:
- $2\text{MnO}_4^- + 10\text{I}^- + 16\text{H}^+ \rightarrow 2\text{Mn}^{2+} + 5\text{I}_2 + 8\text{H}_2\text{O}$
 - $\text{Sn}^{4+} + \text{H}_2\text{O}_2 \rightarrow \text{Sn}^{2+} + 2\text{H}^+ + \text{O}_2$
 - $\text{Cr}_2\text{O}_7^{2-} + 6\text{F}^- + 14\text{H}^+ \rightarrow 2\text{Cr}^{3+} + 3\text{F}_2 + 7\text{H}_2\text{O}$
 - $\text{Cu} + 2\text{H}^+ \rightarrow \text{Cu}^{2+} + \text{H}_2$
 - $2\text{Fe}^{3+} + \text{Sn}^{2+} \rightarrow 2\text{Fe}^{2+} + \text{Sn}^{4+}$
3. (a) Which of the following species could react with $1 \text{ mol L}^{-1} \text{HCl}$ to form hydrogen gas?
- Cu
 - Mg
 - Hg
 - Ag
 - Sn
 - Zn
- (b) From the table of reduction potentials supplied, identify
- a reducing agent which could convert Pb^{2+} to Pb, but not Co^{2+} to Co.
 - an oxidising agent which could convert Cl^- to Cl_2 , but not F^- to F_2 .
 - a reductant which could convert H^+ to H_2 , but not H_2O to H_2 .
 - an oxidant which could convert Ag to Ag^+ , but not Hg to Hg^{2+} .
 - a reductant which could convert acidified MnO_4^- to Mn^{2+} , but not acidified $\text{Cr}_2\text{O}_7^{2-}$ to Cr^{3+} .
4. Predict whether the following disproportionation reactions could occur in aqueous solution:
- copper(I) ion to copper(II) ion and copper metal
 - Iron(II) ion to iron(III) ion and iron metal
 - hydrogen peroxide to water and oxygen gas
 - chlorine to hypochlorous acid and chloride ion
 - manganese dioxide to permanganate ion and manganese(II) ion
5. Predict whether reactions could occur in each of the following. Assume standard conditions.
- Chlorine gas is bubbled through potassium bromide solution.
 - Iron(II) nitrate is mixed with sodium iodide.
 - Aluminium is added to hydrochloric acid.
 - An iron nail is placed in a tin(II) chloride solution.
 - An iron(II) sulfate solution is placed in a nickel container.
 - Hydrogen sulfide is bubbled through an acidified potassium dichromate solution.
 - Chlorine gas is bubbled through an acidified solution of barium nitrate.
 - Chlorine gas is bubbled through an acidified solution of iron(II) bromide.

Electrolysis

An electrolytic redox reaction

Examples

1. In the cell

-
-
-

(b)

(c)

Chemistry Problems

30. $\text{NH}_4\text{NO}_2 \rightarrow \text{N}_2 + 2\text{H}_2\text{O}$
31. $3\text{I}_2 + 6\text{OH}^- \rightarrow \text{IO}_3^- + 5\text{I}^- + 3\text{H}_2\text{O}$
32. $4\text{MnO}_4^- + 4\text{OH}^- \rightarrow 4\text{MnO}_4^{2-} + \text{O}_2 + 2\text{H}_2\text{O}$

Set 19

1. (a) $\text{Zn} + \text{Sn}^{2+} \rightarrow \text{Zn}^{2+} + \text{Sn}$, +0.62V
(b) $3\text{Ag}^+ + \text{Cr} \rightarrow 3\text{Ag} + \text{Cr}^{3+}$, +1.54V
(c) $\text{Hg}^{2+} + \text{Cu} \rightarrow \text{Hg} + \text{Cu}^{2+}$, +0.51V
(d) $\text{Mg} + \text{Cu}^{2+} \rightarrow \text{Mg}^{2+} + \text{Cu}$, +2.70V
(e) $\text{Mg} + 2\text{Ag}^+ \rightarrow \text{Mg}^{2+} + 2\text{Ag}$, +3.16V
(f) $\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{Fe}^{2+} \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O} + 6\text{Fe}^{3+}$, +0.56V
(g) $\text{Cl}_2 + 2\text{I}^- \rightarrow 2\text{Cl}^- + \text{I}_2$, +0.82V
2. (a) Yes, +0.99V (b) No, -0.53V
(c) No, -1.54V (d) No, -0.34V
(e) Yes, +0.62V
3. (a) Mg, Sn, Zn
(b) (i) Sn or Ni
(ii) H_2O_2 or MnO_4^- (these are common oxidants, both must be acidified)
(iii) Pb, Sn, Ni, Co
(iv) $\text{O}_2/4\text{H}^+$
(v) Au, $\text{Cl}^-/\text{H}_2\text{O}$, Cl^-
4. (a) Yes, +0.37V (b) No, -1.21V
(c) Yes, +1.10V (d) No, -0.27V
(e) Yes, +0.63V
5. (a) $\text{Cl}_2 + 2\text{Br}^- \rightarrow 2\text{Cl}^- + \text{Br}_2$, +0.27V
(b) No reaction
(c) $2\text{Al} + 6\text{H}^+ \rightarrow 2\text{Al}^{3+} + 3\text{H}_2$, +1.66V
(d) $\text{Fe} + \text{Sn}^{2+} \rightarrow \text{Fe}^{2+} + \text{Sn}$, +0.30V
(e) No reaction
(f) $\text{Cr}_2\text{O}_7^{2-} + 3\text{H}_2\text{S} + 8\text{H}^+ \rightarrow 2\text{Cr}^{3+} + 3\text{S} + 7\text{H}_2\text{O}$, +1.19V
(g) No reaction
(h) Both Br^- and Fe^{2+} are oxidised

Set 20

	Anode	Cathode	Reaction	Minimum Voltage
1. (a)	I_2	Cu	$\text{Cu}^{2+} + 2\text{I}^- \rightarrow \text{Cu} + \text{I}_2$	0.20V
(b)	Br_2	H_2	$2\text{H}^+ + 2\text{Br}^- \rightarrow \text{H}_2 + \text{Br}_2$	1.09V